Functions on $\mathbb{R}$

$\lim_{x\to a} f(x) = L$: def: $\forall \varepsilon > 0,\ \exists \delta > 0,\ x \in I,\ 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

$f : I \setminus \{a\} \to \mathbb{R}$

e.x. Show: $\lim_{x\to a} (mx+b) = ma+b$

$\forall \varepsilon > 0$. $\exists\ \delta = \frac{\varepsilon}{|m|}$

$0 < |x-a| < \delta \Rightarrow |mx+b-(ma+b)| = |m|\,|x-a| < \varepsilon$.

$\Rightarrow \lim_{x\to a} (mx+b) = ma+b$

e.x. $\lim_{x\to a} x^3 = a^3$

$|x^3 - a^3| = |x-a|\,|x^2+ax+a^2|$

let $M = |a|+1$.

$\forall \varepsilon > 0$. $\exists\ \delta = \min\{1, \frac{\varepsilon}{M^2+aM+a^2}\}$.

$||x|-|a|| < |x-a| < \delta \le 1$.

$|x| \le |a|+1 = M$.

$|x-a| < \delta \Rightarrow |x^3-a^3| = |x-a|\cdot|x^2+ax+a^2|$

$< \delta \cdot |M^2+aM+a^2|$

$< \frac{\varepsilon}{|M^2+aM+a^2|}\,|M^2+aM+a^2| = \varepsilon$.

$\therefore \lim_{x\to a} x^3 = a^3$

rmk: if $f(x) = g(x)$, $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$

e.x. $g(x) = \frac{x^3+x^2-x-1}{x^2-1}$, $\lim_{x\to 1} g(x)$.

$g(x) = \frac{(x^2-1)(x+1)}{(x^2-1)} = x+1$.

$\lim_{x\to 1} g(x) = \lim_{x\to 1} (x+1) = 2$.

The definition can be summarized as

$$\lim_{x\to a} f(x) = L \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I : 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Its negation is

$$\lim_{x\to a} f(x) \neq L \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in I : 0 < |x-a| < \delta \text{ and } |f(x) - L| \geq \varepsilon_0.$$

Thm 3.6: $\lim_{x\to a} f(x) = L \iff$ for every $\{x_n\}_{n=1}^{\infty}$ with $x_n \in I \setminus \{a\}$ and $x_n \to a$, $f(x_n) \to L$

Proof: ① $\lim_{x\to a} f(x) = L \Rightarrow \forall \varepsilon > 0, \exists \delta > 0,\ 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

if $x_n \to a$, $\exists N \in \mathbb{N}$. $n \ge N \Rightarrow 0 < |x_n - a| < \delta$

Hence $|f(x_n) - L| < \varepsilon \Rightarrow f(x_n) \to L$

② suppose $\lim_{x\to a} f(x) \neq L$. there exists $\varepsilon_0 > 0$: $\forall \delta > 0$. $\exists x \in I \setminus \{a\}$.

$|x-a| < \delta$ and $|f(x) - L| \ge \varepsilon_0$.

Choose $\delta < 1$. such that $(a-\delta, a+\delta) \subseteq I$.

For each $n \in \mathbb{N}$. $\delta_n = \min\{\frac{1}{n}, \frac{\delta}{2}\}$

$\Rightarrow |x_n - a| < \delta_n$ $|f(x_n) - L| \ge \varepsilon_0$.

which construct a Sequence $x_n \to a$ but $f(x_n)$ doesn't converge to L.

contrapositive: "$q \to p \iff \neg p \to \neg q$"

Theorem 3.8

I is an open interval containing a. f, g are function on I.

$\lim_{x\to a} f(x) = L$. $\lim_{x\to a} g(x) = M$.

then:

① $\lim_{x\to a} (f+g)(x) = L+M$.

② $\lim_{x\to a} (\alpha f)(x) = \alpha L$.

③ $\lim_{x\to a} (fg)(x) = LM$.

④ $\lim_{x\to a} (\frac{1}{g})(x) = \frac{1}{M}$ given $M \neq 0$

proof: $\{x_n\}$: any sequence in $I \setminus \{a\}$ with $x_n \to a$.

By thm 3.6. $f(x_n) \to L$, $g(x_n) \to M$

By algebraic rule for Sequence:

$f(x_n) + g(x_n) \longrightarrow L+M$.

Since it holds for any sequence. By thm 3.6 $\lim_{x\to a} (f+g)(x) = L+M$.

② ③ ④ can be proved similarly

3.9. Sequeeze theorem

① if $g(x) \le h(x) \le f(x)$ for all $x \in I \setminus \{a\}$

and $\lim_{x\to a} g(x) = \lim_{x\to a} f(x) = L$.

then $\lim_{x\to a} h(x) = L$

② if $|g(x)| \le M$ for all $x \in I \setminus \{a\}$. and $f(x) \to 0$ for $x \to a$.

then $h(x) \to 0$ as $x \to a$.

3.10. Theorem for Comparison

Suppose $a \in \mathbb{R}$. f, g have limits as $x \to a$

and $f(x) \le g(x)$ for all $x \in I \setminus \{a\}$

then $\lim_{x\to a} f(x) \le \lim_{x\to a} g(x)$

One-sided limit and limits at infinity

$\forall \varepsilon > 0$. $\exists\ \delta > 0$, such that $a < x < a+\delta \in I$. $a < x < a+\delta \Rightarrow |f(x) - L| < \varepsilon$.

then we say $f(a^+) := L =: \lim_{x\to a^+} f(x)$

$\forall \varepsilon > 0$. $\exists\ \delta > 0$, such that $a-\delta \in I$. $a-\delta < x < a \Rightarrow |f(x) - L| < \varepsilon$.

then we say $f(a^-) := L =: \lim_{x\to a^-} f(x)$

e.x. Prove: $f(x) = \begin{cases} x+1, & x \ge 0 \\ x-1, & x < 0 \end{cases}$

has one-sided limit at $x=0$ but $\lim_{x\to 0} f(x)$ does not exist

① $\forall \varepsilon > 0\ \exists \delta = \varepsilon$. $0 < x < \delta \Rightarrow |x+1-1| = |x| < \delta = \varepsilon$.

$\therefore \lim_{x\to 0^+} f(x) = 1$.

② $\forall \varepsilon > 0\ \exists \delta = \varepsilon$. $-\delta < x < 0 \Rightarrow |x-1-(-1)| = |x| < \delta = \varepsilon$.

$\therefore \lim_{x\to 0^-} f(x) = -1$.

③ let $x_n = \frac{(-1)^n}{n} \to 0$. $f(x_n) = \frac{(-1)^n}{n} + (-1)^n = (-1)^n(1+\frac{1}{n})$

$\Rightarrow f(x_n)$ does not converge $\Rightarrow \lim_{x\to 0} f(x)$ doesn't exist

e.x. prove: $\lim_{x\to 0^+} \sqrt{x} = 0$

$\forall \varepsilon > 0\ \exists \delta = \varepsilon^2$ $0 < x < \delta \Rightarrow |\sqrt{x}| < |\sqrt{\delta}| < |\sqrt{\varepsilon^2}| < \varepsilon$

$\Rightarrow \lim_{x\to 0^+} \sqrt{x} = 0$

Theorem 3.14.

$\lim_{x\to a} f(x)$ exists and equals to L. if and only if: $L = \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$

proof ① if $f(x) \to L$ as $x \to a$. given $\varepsilon > 0$., choose $\delta > 0$.

$0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

since $a < x < a+\delta$ or $a-\delta < x < a$ both satisfy $0 < |x-a| < \delta$.

Hence both one side limit exists and equals to L

② if $L = \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$

$a < x < a+\delta_1 \Rightarrow |f(x) - L| < \varepsilon$.

and $a-\delta_2 < x < a \Rightarrow |f(x) - L| < \varepsilon$.

Set $\delta = \min\{\delta_1, \delta_2\}$

then $|x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

Thus $f(x) \to L$ as $x \to a$

3.15 Definition

Let $a, L \in \mathbb{R}$, and let f be a real function.

(i) The function $f(x)$ is said to *converge to L as $x \to +\infty$* if and only if there is a $c > 0$ such that $(c, +\infty) \subseteq \mathrm{Dom}(f)$ and, given $\varepsilon > 0$, there is an $M > c$ such that

$$x > M \implies |f(x) - L| < \varepsilon.$$

In this case, we write

$$\lim_{x\to+\infty} f(x) = L \qquad \text{or} \qquad f(x) \longrightarrow L \quad \text{as } x \to +\infty.$$

Similarly, $f(x)$ is said to *converge to L as $x \to -\infty$* if and only if there is a $c > 0$ such that $(-\infty, -c) \subseteq \mathrm{Dom}(f)$ and, given $\varepsilon > 0$, there is an $M < -c$ such that

$$x < M \implies |f(x) - L| < \varepsilon.$$

We write

$$\lim_{x\to-\infty} f(x) = L \qquad \text{or} \qquad f(x) \longrightarrow L \quad \text{as } x \to -\infty.$$

(ii) The function $f(x)$ is said to *converge to $+\infty$ as $x \to a$* if and only if there is an open interval I containing a such that $I \setminus \{a\} \subseteq \mathrm{Dom}(f)$ and, given $M \in \mathbb{R}$, there is a $\delta > 0$ such that

$$0 < |x-a| < \delta \implies f(x) > M.$$

(1) $f(x)$ converges to L as $x \to \infty$.

if and only if: there is $c > 0$ and.

$(c, +\infty) \subseteq \mathrm{Dom}(f)$. and. given $\varepsilon > 0$.

$\exists\ M > c$. $x > M \Rightarrow |f(x) - L| < \varepsilon$.

(2). $f(x)$ converges to ∞ as $x \to a$.

$\Leftrightarrow \forall M > 0$. $\exists\ \delta$ $0 < |x-a| < \delta \Rightarrow f(x) > M$

We write

$$\lim_{x\to a} f(x) = +\infty \qquad \text{or} \qquad f(x) \longrightarrow +\infty \quad \text{as } x \to a.$$

Similarly, $f(x)$ is said to *converge to $-\infty$ as $x \to a$* if and only if there is an open interval I containing a such that $I \setminus \{a\} \subseteq \mathrm{Dom}(f)$ and, given $M \in \mathbb{R}$, there is a $\delta > 0$ such that

$$0 < |x-a| < \delta \implies f(x) < M.$$

We write

$$\lim_{x\to a} f(x) = -\infty \qquad \text{or} \qquad f(x) \longrightarrow -\infty \quad \text{as } x \to a.$$

Other infinite-limit combinations

The remaining combinations are defined analogously: $f(x)$ may tend to $+\infty$ or $-\infty$ as $x \to a+$ or $x \to a-$, and it may tend to $+\infty$ or $-\infty$ as $x \to +\infty$ or $x \to -\infty$.

e.x. prove: $1/x \to 0$ as $x \to \infty$

$\forall \varepsilon > 0$ $\exists\ M = \frac{1}{\varepsilon}$. $x > M \Rightarrow |\frac{1}{x} - 0| = |\frac{1}{x}| < \frac{1}{|M|} = \varepsilon$.

e.x. $f(x) = \frac{x+2}{2x^2-3x+1}$

prove $\lim_{x\to 1^-} f(x) = -\infty$.

$f(x) = \frac{x+2}{(2x-1)(x-1)}$.

$\exists\ \delta \in (0,1)$. $1-\delta < x < 1 \Rightarrow \frac{3}{M} < 2x^2-3x+1 < 0$.

$\frac{1}{2x^2-3x+1} > \frac{3}{M}$

Continuity

Def: f is continuous at a if: $\forall \varepsilon > 0$. $\exists\ \delta > 0$ such that $x \in I$, $|x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$

f is continuous on I if it is continuous at every point of I.

$\lim_{x\to a} f(x) = f(a)$ ⟺ f is continuous at a

Endpoint convention. If $f : [a,b] \to \mathbb{R}$, continuity at the endpoints is understood relative to the domain:

$$\lim_{x\to a^+} f(x) = f(a), \qquad \lim_{x\to b^-} f(x) = f(b).$$

Theorem 3.21. Sequential Characterization

f is cont. at $a \iff$ every $(x_n) \subseteq I$ with $x_n \to a$ satisfies: $f(x_n) \to f(a)$.

proof: ① f is cont. at $a \Rightarrow \forall \varepsilon > 0\ \exists \delta > 0$. $0 < |x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$.

since $x_n \to a$. Choose this δ. $\exists N$. $n > N \Rightarrow |x_n - a| < \delta \Rightarrow |f(x_n) - f(a)| < \varepsilon$

$\Rightarrow f(x_n) \to f(a)$ as $n \to \infty$

② Assume. f is not cont. at a.

$\exists\ \varepsilon_0 > 0$. $\forall \delta > 0$. $0 < |x-a| < \delta \Rightarrow |f(x) - f(a)| > \varepsilon$.

for every $n \in \mathbb{N}$. Let $\delta = \frac{1}{n}$. $|x_n - a| < \frac{1}{n}$. Satisfies $x_n \to a$.

But $|f(x_n) - f(a)| > \varepsilon_0$. $f(x_n)$ does not converge to $f(a)$

$\neg p \to \neg q \iff q \to p$

Theorem: Algebra of cont. functions

① $f, g : I \to \mathbb{R}$ be cont. at $a \in I$

then: $f+g$. $f \cdot g$. $\frac{f}{g}$ $(g(a) \neq 0)$ are Cont. at a.

② if f, g cont. on I

then: $f+g$. $f \cdot g$. $\frac{f}{g}$ $(g(a) \neq 0)$ are Cont. on I.

proof: $\lim_{x\to a} (f(x) + g(x)) = f(a) + g(a) = (f+g)(a)$

Def: Composition

$f : A \to \mathbb{R}$. $g : B \to \mathbb{R}$. assume: $f(A) \subseteq B$.

The Composition of g with f: $g \circ f : A \to \mathbb{R}$. $(g \circ f)(x) = g(f(x))$

Theorem: Continuity & Composition

① $f : A \to \mathbb{R}$. $g : B \to \mathbb{R}$. if f is cont at $a \in A$, g is cont. at $f(a) \in B$.

the $g \circ f$ is Cont. at a.

② if f is cont. on A & g is cont. on B. then $g \circ f$ is cont. on A.

proof: (let $x_n \in A$ and $x_n \to a$. cont. of f $f(x_n) \to f(a)$.

since $f(A) \subseteq B$, $f(x_n) \in B$ $\forall n$ cont. of g $g(f(x_n)) \to g(f(a))$.

Hence. $g \circ f$ is cont. at a

Def: Bounded Function

$E \subseteq \mathbb{R}$, $f : E \to \mathbb{R}$, f is bounded if $f(E) = \{f(x) : x \in E\}$ is bounded

Equivalently: there is an $M > 0$. such that $|f(x)| \le M$.

in this case. f is said to be dominated by M on E

e.x. $f : [0,1] \to \mathbb{R}$, $f(x) = 5x^3 + 4x^2$. For $x \in [0,1]$.

$|f(x)| \le 5|x|^3 + 4|x|^2 \le 5+4 = 9$.

$\Rightarrow f$ is bounded on $[0,1]$. (f is dominated by 9 on $[0,1]$).

Theorem: Extreme Value Theorem

① $I = [a,b]$. $f : I \to \mathbb{R}$ be cont. Then f is bounded on I.

② Moreover. if $M = \sup_{x\in I} f(x)$. $m = \inf_{x\in I} f(x)$.

then there are points $x_M, x_m \in I$:

$f(x_M) = M$. $f(x_m) = m$

proof ① suppose f is unbounded. For every $n \in \mathbb{N}$ choose $x_n \in I$ such that. $|f(x_n)| > n$.

since $x_n \in I$. (x_n) is bounded. —BW→ (x_n) has a Convergent subseq. (x_{n_k}).

Say $x_{n_k} \to c$., since f is continuous. $f(x_{n_k}) \to f(c) \in \mathbb{R}$.

$|f(x_{n_k})| > n \to \infty$. Hence a Contradiction

② Let $M = \sup_{x\in I} f(x)$, Choose $x_n \in I$, $f(x_n) \to M$. A subsequence x_{n_k} Converges to Some $x_M \in I$.

$\lim_{k\to\infty} f(x_{n_k}) = f(x_M)$ & $\lim_{k\to\infty} f(x_{n_k}) = M$ $\Rightarrow$ $f(x_M) = M$

x_m can be attained Similarly

3.27 Remark (why both hypotheses matter). Neither compactness hypothesis can be omitted:

- $f(x) = 1/x$ is continuous but unbounded on the bounded, nonclosed interval $(0,1)$;
- $f(x) = x$ is continuous and bounded on $(0,1)$, but $\sup f((0,1)) = 1$ is not attained;
- $f(x) = x$ is continuous but unbounded on the closed, unbounded interval $[0, +\infty)$.

{ ① closed. bounded interval $I = [a,b]$
② continuous $f : I \to \mathbb{R}$

Lemma: Persistence of positivity

$a < b$. $f : [a,b) \to \mathbb{R}$. if f is Continuous at x_0 and $f(x_0) > 0$.

then there are an $\varepsilon > 0$ and an $x_1 \in (x_0, b)$ such that $f(x) > \varepsilon$. ($x \in [x_0, x_1]$)

proof:

Set $\varepsilon = f(x_0)/2 > 0$. since f is cont. at x_0. choose $\delta > 0$. such that: $|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \frac{f(x_0)}{2}$

Choose $x_1 \in (x_0, b)$. close to x_0 that $x_1 - x_0 < \delta \Rightarrow \forall x \in [x_0, x_1]$ $|f(x) - f(x_0)| < \frac{f(x_0)}{2}$

$\Rightarrow f(x) > f(x_0) - \frac{f(x_0)}{2} = \frac{f(x_0)}{2} = \varepsilon$.

Intermediate Value Theorem

$f : [a,b]$ be continuous. if y_0 lies Strictly between $f(a)$ and $f(b)$. then there is $x_0 \in (a,b)$. $f(x_0) = y_0$.

proof: Assume: $f(a) < y_0 < f(b)$. Set: $E := \{x \in [a,b] : f(x) < y_0\}$., $x_0 := \sup E$

E is nonempty & bounded. Choose $x_n \in E$ with $x_n \to x_0$, since $E \subseteq [a,b]$, $x_0 \in [a,b]$

and since f is cont. $f(x_0) = \lim_{n\to\infty} f(x_n) \le y_0$., also shows $x_0 < b$

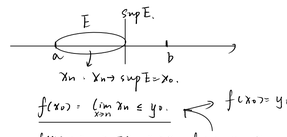
If $f(x_0) < y_0$., let $h(x) := y_0 - f(x)$, $h(x)$ is cont. and positive at x_0.

By Lemma. $\exists\ \varepsilon > 0$. $x_1 \in (x_0, b)$ $h(x) > \varepsilon$ ($x_0 \le x \le x_1$).

$\Rightarrow h(x) > 0 \Rightarrow f(x_1) < y_0$., since $x_0 = \sup E$. & $x_1 \in E$ & $x_1 > x_0$. a Contradiction.

$\therefore f(x_0) = y_0$

Together with Extreme value theorem. IVT shows that $f([a,b])$ is a closed interval



$f(x_0) = \lim_{x\to x_n} x_n \le y_0$.

$f(x_0) < y_0 \Rightarrow \exists x_1, x_1 > x_0$. $f(x_1) < y_0$ $\times$.

$\therefore$

$f(x_0) = y_0$

When continuity fail?

f is Cont. at $a \iff \lim_{\substack{x\to a \\ a\in I}} f(x) = f(a)$. fail: { 1. $\lim_{x\to a} f(x)$ does not exist; 2. $\lim_{x\to a} f(x) \neq f(a)$.

e.x. $f(x) = \begin{cases} |x|/x, & x \neq 0 \\ 1, & x = 0 \end{cases}$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} 1 = 1$. $\lim_{x\to 0^-} f(x) = -1$.

$\Rightarrow \lim_{x\to 0} f(x)$ does not exist $\Rightarrow f$ is not cont. at a

e.x. $f(x) = \begin{cases} \sin(1/x), & x \neq 0 \\ 1, & x = 0 \end{cases}$

let. $x_n = \frac{1}{2\pi n} \to 0$. $f(x_n) = \sin(2\pi n) = 0$. $\forall n$.

$y_n = \frac{1}{2n\pi + \pi/2} \to 0$. $f(x_n) = 1$. $\forall n$.

$\lim_{n\to\infty} f(x_n) \neq \lim_{n\to\infty} f(y_n) \Rightarrow$ limit doesn't exist

3.32 Example (Dirichlet's function). Let

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

The function D is nowhere continuous. Indeed, for every $a \in \mathbb{R}$, the density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ supplies sequences $x_n \in \mathbb{Q}$ and $y_n \notin \mathbb{Q}$ with $x_n \to a$ and $y_n \to a$. But

$$D(x_n) = 1, \qquad D(y_n) = 0,$$

so both sequences cannot have images converging to $D(a)$.

Def: Uniform Continuity

f is uniformly continuous on I ⟺ $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in I$. $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.

Lemma: if $f : I \to \mathbb{R}$ is uniformly cont. on I. then f is continuous on I.

proof: Fix $a \in I$. $\forall \varepsilon > 0$. $\exists\ \delta > 0$. $|x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$. $\Rightarrow f(x)$ is cont. at a

$\Rightarrow$ consequently $f(x)$ is Cont. on I

e.x. $f(x) = x^2$ is uniformly continuous on $(0,1)$

proof: $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$

$|x^2 - y^2| = |x+y|\,|x-y| < \varepsilon$

$0 < x, y < 1 \Rightarrow 0 < x+y < 2$.

take $\delta = \frac{\varepsilon}{2}$. $|x+y|\,|x-y| < 2|x-y| < 2\cdot\frac{\varepsilon}{2} = \varepsilon$

Hence $f(x)$ is uniformly continuous on $(0,1)$

e.x. $f(x)$ is not uniformly continuous on $\mathbb{R}$

$\exists\ \varepsilon_0 > 0$. $\forall \delta > 0$ $\exists^{x,y\in I}$ $0 < |x-y| < \delta \Rightarrow |f(x) - f(y)| \ge \varepsilon_0$, $\Rightarrow |x-y|\,|x+y| > \varepsilon_0$

equivalently or $\delta = \frac{1}{n}$, $n \in \mathbb{N}$ $\exists\ x_n, y_n \in I$. $|x_n - y_n| < \frac{1}{n} \Rightarrow |f(x_n) - f(y_n)| > \varepsilon_0$ → Unbounded

let. $x_n = n$. $y_n = n + \frac{1}{n}$ $|x-y| = \frac{1}{n} \to 0$, $|f(x_n) - f(y_n)| \cdot |x_n - y_n| \cdot \frac{1}{n}(2n + \frac{1}{n}) > 2$.

3.38 Cauchy Sequence (Lemma)

$f : I \to \mathbb{R}$ is uniformly continuous. If $(x_n) \subseteq I$ is a Cauchy Sequence

then $f(x_n)$ is a cauchy sequence

proof: $\forall \varepsilon > 0$. $\exists\ \delta > 0$. $0 < |x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$

since $(x_n) \subseteq I$ is cauchy sequence for this δ. $\exists N$. $m, n \ge N \Rightarrow |x_m - x_n| < \delta \Rightarrow |f(x_n) - f(x_m)| < \varepsilon$

$\Rightarrow f(x_n)$ is also a cauchy sequence

Remark: we must have (x_n) is Cauchy

Theorem (Heine Cantor)

Let $I = [a,b]$ be closed & bounded interval. if: $f : I \to \mathbb{R}$ is continuous then f is uniformly continuous on I.

proof: if not: $\exists\ \varepsilon_0 > 0$. $\forall\ \frac{1}{n}$. $n \in \mathbb{N}$. $\exists\ x_n, y_n \in I$. $|x_n - y_n| < \frac{1}{n} \Rightarrow |f(x_n) - f(y_n)| > \varepsilon_0$.

First: $(x_n) \subseteq I = [a,b] \xrightarrow{\text{B.W.}} \exists$ Subsequence (x_{n_k}) is Convergent

Say $(x_{n_k}) \to \alpha$.

Then: $(y_{n_k}) \subseteq I$. By BW theorem, $\exists$ Subsequence of (y_{n_k}) is convergent.

Say $(y_{n_{k_j}}) \to \beta$.

subsequence of (x_{n_k}): $(x_{n_{k_j}})$ also Convergent to α.

$|x_{n_{k_j}} - y_{n_{k_j}}| \longrightarrow |\alpha - \beta|$ as $j \to \infty$

$\Rightarrow \frac{1}{n} > |x_{n_{k_j}} - y_{n_{k_j}}| \to |\alpha - \beta|$

$\Rightarrow \alpha = \beta \in [a,b]$.

$\varepsilon_0 < |f(x_{n_{k_j}}) - f(y_{n_{k_j}})| \le 1 \cdot |f(x_{n_{k_j}}) - f(\beta)| \to 0$. ($\xrightarrow{f(\alpha)}$ $1 \cdot |f(y_{n_{k_j}}) - f(\beta)|$)

contradiction

Theorem: Continuous extension Criterion

$a < b$. $f : (a,b) \to \mathbb{R}$.

f is uniformly continuous on (a,b) ⟺ there is a continuous extension:

there is Continuous function $g : [a,b] \to \mathbb{R}$. : $g(x) = f(x)$, $x \in (a,b)$

proof: ① Since g is continuous on $[a,b]$. g is uniformly Continuous on $[a,b]$.

$(a,b) \subseteq [a,b]$. g is uniformly cont. on (a,b)

② (let $f : (a,b) \to \mathbb{R}$. be uniformly cont. on (a,b).

Need to find a $g : [a,b] \to \mathbb{R}$. cont. on $[a,b]$.

(1). Define $g(a)$, $g(b)$:

For any sequence $(x_n) \subseteq (a,b)$, $x_n \to a$.

x_n is cauchy. f is uniformly cont. on (a,b).

$\Rightarrow f(x_n)$ is also a cauchy sequence

$g(a) = \lim_{n\to\infty} f(x_n) =: \alpha$.

$(y_n) \subseteq (a,b)$, $y_n \to a$. Claim: $\lim_{n\to\infty} f(y_n) = \lim_{n\to\infty} f(x_n) = \alpha$, So $g(a)$ is well-defined

$|x_n - y_n| < |x_n - a| + |y_n - a| \to 0$.

$\Rightarrow |f(x_n) - f(y_n)| \to 0$ $\Rightarrow \lim f(x_n) = \lim f(y_n)$

$g(a) = \lim_{n\to\infty} f(x_n) = \lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x)$.

$g(x)$ is continuous at a

Similarly we define $g(b) = \lim_{n\to\infty} f(z_n)$, where $z_n \in (a,b)$ $z_n \to b$. $\Rightarrow g$ is cont. at b

$g(x) = f(x)$. $\forall x \in (a,b)$

$\Rightarrow g : [a,b] \to \mathbb{R}$ is continuous

$f : (a,b) \to \mathbb{R}$ is uniformly cont. ⟺ $\lim_{x\to a^+} f(x) = \alpha$. $\lim_{x\to b^-} f(x) = \beta$

e.x. $f(x) = \frac{x-1}{\log x}$. $x \in (0,1)$ is uniformly cont. on $(0,1)$.

proof: $\lim_{x\to 0^+} \frac{x-1}{\log x} = 0$ $\lim_{x\to 1^-} \frac{x-1}{\log x} = 1$.

let $g(x) = \begin{cases} 0, & x = 0 \\ \frac{x-1}{\log x}, & x \in (0,1) \\ 1, & x = 1 \end{cases}$

$g(x)$ is continuous on $[0,1] \Rightarrow f$ is uniformly cont. on $(0,1)$